COUNTABLE-CHAIN-CONDITION FORCING FOR COMPLEMENTARY UNCOUNTABLY CHROMATIC GRAPHS WITH COUNTABLE PRODUCT

SOUKUP, LAJOS

Editorial note. Original mathematical body, original TikZ diagrams and embedded bibliography retained without mathematical edits, additional assumptions or independent certification.

Editorial note. Native declaration1031–1037 retained: author used ChatGPT for grammar, typography and misprints, states all mathematical arguments were created by author and takes full publication responsibility. No further provenance investigation performed.

Editorial note. The exact size obstruction, diamond construction and ccc forcing theorem are separate native main statements; no direct consequence or standard input lemma separately counted.

Allowed standard prerequisites (editorial).

- Categorical graph products, uncountable chromatic number, Countryman orders and ccc forcing. Original complement/product definitions, minimal Countryman-line standard inputs, finite synchronized forcing predicates and all freshness/amalgamation lemmas retained.. Entire native mathematical body, both TikZ figures and embedded bibliography preserved; distinct size obstruction, diamond construction and forcing-existence statements retained exactly.

1. Introduction

All graphs in this note are simple and undirected. If $G_j = (V_j, E_j)$ for $j < 2$, their (categorical) product is

$$G_0 \times G_1 = (V_0 \times V_1, \{\{(a_0, a_1), (b_0, b_1)\} : \{a_0, b_0\} \in E_0 \text{ and } \{a_1, b_1\} \in E_1\}) .$$

For a graph $G = (V, E)$, write $\overline{G} = (V, [V]^2 \setminus E)$. The study of inequalities relating a graph and its complement goes back to Nordhaus and Gaddum [5].

Pulling back a coloring along either coordinate projection gives

$$\mathrm{chr}(G_0 \times G_1) \leq \min\{\mathrm{chr}(G_0), \mathrm{chr}(G_1)\}.$$

Hedetniemi conjectured in 1966 that equality holds for finite graphs [3], but Shitov disproved the conjecture [10]; see [8, 12] for surveys. For infinite graphs, equality can fail much more dramatically. Hajnal showed that for every infinite cardinal κ there are graphs G_0, G_1 such that

$$\mathrm{chr}(G_0) = \mathrm{chr}(G_1) = \kappa^+ \quad \text{and} \quad \mathrm{chr}(G_0 \times G_1) = \kappa.$$

Thus two ω_1-chromatic graphs may have a countably chromatic product [2]. Soukup obtained two ω_2-chromatic graphs with countably chromatic product in a model of

Date: arxiv:2610.08027v1; CC BY 4.0.

GCH [11]. Rinot later showed that in the constructible universe, for every infinite cardinal κ, there are graphs $\langle G_i : i < \kappa^+ \rangle$ on κ^+ such that

$$\mathrm{chr}(G_i) = \kappa^+ \quad \text{and} \quad \mathrm{chr}(G_i \times G_j) = \omega \qquad (i < j < \kappa^+).$$

Their edge sets may moreover be chosen pairwise disjoint [7]. Consequently, for $i \neq j$, the complement of $G_i \cup G_j$ contains every G_k with $k \notin \{i, j\}$, and hence has chromatic number κ^+. This suggests looking at the opposite extreme, where two edge-disjoint graphs cover every pair—that is, where the factors are a graph and its complement.

The central question of this paper is how large the chromatic numbers of G and $\overline{G}$ can be while their product remains countably chromatic. In particular, can one have

$$(*) \qquad \mathrm{chr}(G), \mathrm{chr}(\overline{G}) > \omega \quad \text{and} \quad \mathrm{chr}(G \times \overline{G}) = \omega.$$

The complementary requirement imposes a strong restriction.

Theorem 1.1. *If G and $\overline{G}$ are uncountably chromatic and $G \times \overline{G}$ is countably chromatic, then $|V(G)| = \omega_1$.*

Thus ω_1 is the only possible cardinality of such an example. Despite these obstructions, this remaining possibility can be realized.

Theorem 1.2. *Assume $\diamondsuit$. There is a graph G on ω_1 such that*

$$\mathrm{chr}(G) = \mathrm{chr}(\overline{G}) = \omega_1 \quad \textit{and} \quad \mathrm{chr}(G \times \overline{G}) = \omega.$$

The proof uses two suitably chosen minimal Countryman lines.

Theorem 1.3. *There is a c.c.c. partial order P of cardinality ω_1 such that every P-generic extension contains a graph G on ω_1 for which*

$$\mathrm{chr}(G) = \mathrm{chr}(\overline{G}) = \omega_1 \quad \textit{and} \quad \mathrm{chr}(G \times \overline{G}) = \omega.$$

Problem 1.4. Does ZFC prove that there is a graph G on ω_1 satisfying $(*)$?

2. The negative results

The proof of Theorem 1.1 rests on the following lemma, based on an observation of Hajnal [2].

Lemma 2.1. *There is no graph G such that*

$$\mathrm{chr}(G), \mathrm{chr}(\overline{G}) \geq \omega_2 \quad \textit{and} \quad \mathrm{chr}(G \times \overline{G}) \leq \omega.$$

Since no proof of Hajnal's observation is given there, we include the full argument for completeness.

Proof of Lemma 2.1. Suppose otherwise. Let $c : V(G) \times V(G) \to \omega$ be a proper coloring of $G \times \overline{G}$, and choose $X \in [V(G)]^{\omega_1}$. We first show that $G \restriction X$ is countably chromatic. Otherwise, for every $y \in V(G)$ the map

$$x \longmapsto c(x, y), \qquad x \in X,$$

fails to be a proper coloring of $G \restriction X$. Choose an edge $e_y = \{a_y, b_y\} \in E(G) \cap [X]^2$ and an $n_y < \omega$ such that

$$c(a_y, y) = c(b_y, y) = n_y.$$

Then

$$y \longmapsto (e_y, n_y)$$

is a proper coloring of $\overline{G}$: if $\{y, y'\} \in E(\overline{G})$ and $(e_y, n_y) = (e_{y'}, n_{y'})$, writing $e_y = e_{y'} = \{a, b\}$ and $n_y = n_{y'} = n$ would make (a, y) and (b, y') adjacent vertices of $G \times \overline{G}$ with the same c-color. Since

$$|(E(G) \cap [X]^2) \times \omega| \leq \omega_1,$$

this contradicts $\mathrm{chr}(\overline{G}) \geq \omega_2$. Hence $G \restriction X$ is countably chromatic. Interchanging the two coordinates and replacing G by $\overline{G}$ shows that $\overline{G} \restriction X$ is countably chromatic as well. Let $c_0, c_1 : X \to \omega$ be proper colorings of these two graphs. The map

$$x \longmapsto \big(c_0(x), c_1(x)\big)$$

is injective, because every pair of distinct elements of X is an edge of exactly one of the two complementary graphs. This contradicts $|X| = \omega_1$. □

Proof of Theorem 1.1. By Lemma 2.1, after interchanging G and $\overline{G}$ if necessary, we have $\mathrm{chr}(G) = \omega_1$. Suppose that $|V(G)| > \omega_1$, and let $d : V(G) \to \omega_1$ be a proper coloring. Some color class C of d is uncountable, since otherwise $V(G)$ would be the union of ω_1 countable sets. Thus C is a clique in $\overline{G}$.

Let $c : V(G) \times V(G) \to \omega$ be a proper coloring of $G \times \overline{G}$. For every $x \in V(G)$, choose $n_x < \omega$ such that

$$A_x = \{y \in C : c(x, y) = n_x\}$$

is uncountable. The map $x \mapsto n_x$ is a proper coloring of G: if $\{x, x'\} \in E(G)$ and $n_x = n_{x'}$, choose distinct $y \in A_x$ and $y' \in A_{x'}$. Then (x, y) and (x', y') are adjacent in $G \times \overline{G}$ and have the same c-color, a contradiction. Hence G is countably chromatic, again a contradiction. Therefore $|V(G)| = \omega_1$. □

3. A construction from ♢

Recall that a linear order is *scattered* if it contains no copy of the rationals, and it is *σ-scattered* if it is the union of countably many scattered suborders. A non-σ-scattered linear order L is *minimal* if L embeds into each of its non-σ-scattered suborders. A *Countryman line* is a linear order $L = (X, <_L)$ of cardinality ω_1 such that the coordinatewise order on X^2 is the union of countably many chains.

We use the following consequence of a theorem of Shalev [9, Corollary 1.2 and Remark 1.3]: ♢ implies that there is a family of 2^{ω_1} pairwise non-embeddable Countryman lines, each minimal with respect to being non-σ-scattered.

Proof of Theorem 1.2. Choose two members L_0, L_1 of Shalev's family such that

$$L_0 \not\hookrightarrow L_1 \quad \text{and} \quad L_0 \not\hookrightarrow L_1^*. \tag{3.1}$$

Here L^* denotes the reverse of L. To see that this choice is possible, fix L_0. There is at most one member M of the family for which L_0 embeds into M^*. Indeed, if both M and N had this property, then L_0^* would embed into both M and N. Its images are non-σ-scattered, so the minimality of M and N would give embeddings

$$M \longrightarrow L_0^* \longrightarrow N,$$

contrary to the pairwise non-embeddability of the family. We may therefore choose L_1 outside this possible exception; the first part of (3.1) follows directly from pairwise non-embeddability.

Relabel the two orders so that they have the same underlying set X of cardinality ω_1, and write them as $(X, <_0)$ and $(X, <_1)$. Define a graph G on X by declaring distinct x, y adjacent precisely when the two orders agree on their orientation:

$$\{x, y\} \in E(G) \quad \Longleftrightarrow \quad (x <_0 y \Longleftrightarrow x <_1 y). \tag{3.2}$$

We first check the chromatic numbers of the two complementary graphs. If G had a countable proper coloring, write $X = \bigcup_{n<\omega} A_n$ with every A_n independent.

Some $L_0 \restriction A_n$ is non-σ-scattered, since a countable union of σ-scattered orders is σ-scattered. Minimality gives an embedding $L_0 \longrightarrow L_0 \restriction A_n$. On A_n the two orders disagree on every pair, so the identity embeds $L_0 \restriction A_n$ into L_1^*. Their composition contradicts (3.1). Thus $\mathrm{chr}(G) = \omega_1$. If $\overline{G}$ had a countable proper coloring, the same argument would use a color class on which the two orders agree and give an embedding $L_0 \longrightarrow L_1$. Consequently,

$$\mathrm{chr}(G) = \mathrm{chr}(\overline{G}) = \omega_1. \tag{3.3}$$

For $j < 2$, choose a cover

$$X^2 = \bigcup_{n<\omega} C_n^j$$

in which every C_n^j is a chain in the coordinatewise order induced by $<_j$. Choose $k_j(x,y) < \omega$ with $(x,y) \in C^j_{k_j(x,y)}$, and put

$$f(x,y) = \big(k_0(x,y), k_1(x,y)\big) \in \omega^2.$$

We claim that f is a proper countable coloring of $G \times \overline{G}$. Suppose that $f(x,y) = f(x',y')$, where $x \neq x'$ and $y \neq y'$. For each $j < 2$, the points (x,y) and (x',y') lie in one coordinatewise chain. Hence

$$x <_j x' \quad \Longleftrightarrow \quad y <_j y'.$$

It follows that the two orders agree on $\{x,x'\}$ exactly when they agree on $\{y,y'\}$. By (3.2), one of these pairs cannot be an edge of G and the other an edge of $\overline{G}$. This proves the claim and gives $\mathrm{chr}(G \times \overline{G}) \leq \omega$.

For completeness, we show that the last inequality is an equality. Let $W = W_0 \dot\cup W_1$ be countably infinite, disjoint from X, with both pieces infinite. Extend G to a graph $\widehat{G}$ on $X \cup W$ by adding all pairs from $[W_0]^2$ and no other pairs involving W. The restrictions to X show that both $\widehat{G}$ and its complement still have chromatic number ω_1.

Fix an injection $e : W \to \omega$. Using three disjoint countable palettes (whose union we identify with ω), extend the product coloring by

$$F(u,v) = \begin{cases} (0, f(u,v)), & u,v \in X, \\ (1, e(u)), & u \in W, \\ (2, e(v)), & u \in X, v \in W. \end{cases} \tag{3.4}$$

Two vertices receiving the same color of the second kind have the same first coordinate, and two receiving the same color of the third kind have the same second coordinate; neither pair can be adjacent in the categorical product. Thus (3.4) properly colors $\widehat{G} \times \overline{\widehat{G}}$ with countably many colors. On the other hand, pairing the members of the clique W_0 bijectively with the members of the $\widehat{G}$-independent set W_1 produces a copy of K_ω in the product. Therefore

$$\mathrm{chr}(\widehat{G} \times \overline{\widehat{G}}) = \omega.$$

Finally, relabel $X \cup W$ as ω_1. □

The preceding argument cannot be carried out in ZFC as stated: it requires suitably chosen minimal Countryman lines, and ZFC does not even prove that a minimal Countryman line exists. Indeed, Moore proved that it is consistent with CH that ω_1 and ω_1^* are the only minimal uncountable linear orders [4]; see also [1, Theorem 1.10]. Since every uncountable suborder of a Countryman line is again a Countryman line, that model has no minimal Countryman lines. The forcing construction below shows that the graph-theoretic phenomenon is nevertheless not intrinsically tied to $\diamondsuit$. We define a c.c.c. forcing of cardinality ω_1 that adds the

required graph over any ground model. In particular, forcing over a model of ¬CH shows that the existence of such a graph is consistent with ¬CH. The argument also isolates the finite extension and amalgamation properties underlying the construction.

4. The forcing construction

Definition 4.1. A *precondition* is a triple

$$p = (A^p, h^p, f^p),$$

where $A^p \in [\omega_1]^{<\omega}$, $h^p : [A^p]^2 \to 2$, and $f^p : A^p \times A^p \to \omega$. We order preconditions by coordinatewise extension:

$$p \le q \quad\Longleftrightarrow\quad A^p \supseteq A^q, \quad h^p \supseteq h^q, \quad f^p \supseteq f^q.$$

Put

$$E^p_i = (h^p)^{-1}\{i\}, \qquad H^p_i = (A^p, E^p_i) \quad (i < 2).$$

Thus H^p_0 and H^p_1 are complementary graphs on A^p. For $e < 2$ define

$$f^p_e(a,b) = \begin{cases} f^p(a,b), & e = 0, \\ f^p(b,a), & e = 1. \end{cases}$$

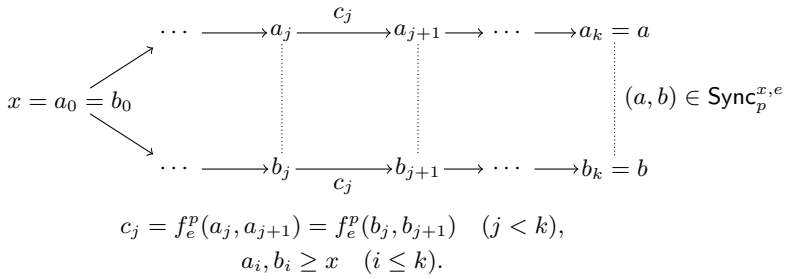
Synchronized pairs. Fix $x \in A^p$ and $e < 2$. Let $\mathsf{Sync}^{x,e}_p \subseteq A^p \times A^p$ consist of all pairs (a,b) for which there are $k < \omega$ and two sequences

$$a_0, \dots, a_k \quad \text{and} \quad b_0, \dots, b_k$$

such that

(1) $a_0 = b_0 = x$, $a_k = a$, and $b_k = b$;
(2) $a_j, b_j \ge x$ for every $j \le k$;
(3) $f^p_e(a_j, a_{j+1}) = f^p_e(b_j, b_{j+1})$ for every $j < k$.

We call such a pair of sequences *synchronized walks* witnessing $(a,b) \in \mathsf{Sync}^{x,e}_p$. The case $k = 0$ is allowed. Notice also that, whenever $x \le a$, the identical one-step walks show $(a,a) \in \mathsf{Sync}^{x,e}_p$.



$$c_j = f^p_e(a_j, a_{j+1}) = f^p_e(b_j, b_{j+1}) \quad (j < k),$$
$$a_i, b_i \ge x \quad (i \le k).$$

FIGURE 1. Synchronized walks witnessing $(a,b) \in \mathsf{Sync}^{x,e}_p$. Corresponding steps have the same f^p_e-color, and all vertices of the walks are at least x.

Remark 4.2. Equivalently, $\mathsf{Sync}^{x,e}_p$ is the least relation $R \subseteq A^p \times A^p$ satisfying the following two rules:

(R0) $(x,x) \in R$;
(R1) if $(a,b) \in R$ and $a', b' \in A^p$ satisfy

$$a', b' \ge x \quad \text{and} \quad f^p_e(a,a') = f^p_e(b,b'),$$

then $(a',b') \in R$.

Indeed, iterating (R1) produces exactly the synchronized walks in the definition above.

Definition 4.3 (Good preconditions)**.** A precondition p is *good* if the following two requirements hold.

(G1) f^p is a proper coloring of $H^p_0 \times H^p_1$. Equivalently, whenever $a_0 \neq b_0$, $a_1 \neq b_1$,

$$h^p(\{a_0, b_0\}) = 0 \quad \text{and} \quad h^p(\{a_1, b_1\}) = 1,$$

then

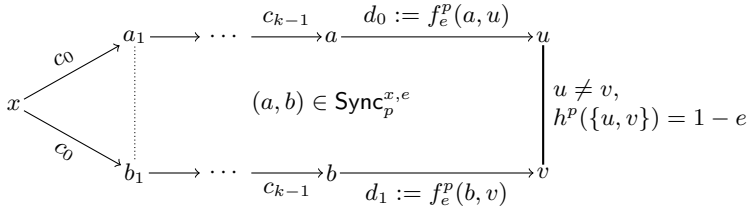
$$f^p(a_0, a_1) \neq f^p(b_0, b_1).$$

(G2) If $e < 2$, $x, a, b, u, v \in A^p$, $u \neq v$,

$$(a, b) \in \mathsf{Sync}^{x,e}_p, \qquad \min\{u, v\} < x, \qquad h^p(\{u, v\}) = 1 - e,$$

then

$$f^p_e(a, u) \neq f^p_e(b, v).$$



$$\min\{u, v\} < x \implies d_0 \neq d_1 \qquad \text{(G2)}$$

FIGURE 2. Requirement (G2). If $(a, b) \in \mathsf{Sync}^{x,e}_p$, $\min\{u, v\} < x$, and $h^p(\{u, v\}) = 1 - e$, then the terminal colors $d_0 = f^p_e(a, u)$ and $d_1 = f^p_e(b, v)$ are distinct.

Let P be the poset of all good preconditions, ordered by coordinatewise extension. The next two lemmas establish the extension and amalgamation properties needed for P.

Lemma 4.4 (Fresh extensions)**.** *Let p be good, let $A^p \subseteq A \in [\omega_1]^{<\omega}$, and let $h : [A]^2 \to 2$ extend h^p. Then there is a map $f : A^2 \to \omega$ such that (A, h, f) is a good extension of p.*

Proof. Extend f^p to $f : A^2 \to \omega$ by giving the members of $A^2 \setminus (A^p)^2$ pairwise distinct values outside $\operatorname{ran}(f^p)$, and set $q = (A, h, f)$. Call the ordered pairs in $(A^p)^2$ *old* and those in $A^2 \setminus (A^p)^2$ *new*. The choice of f gives the following *freshness property*:

(F) $$s \in A^2, \quad t \in A^2 \setminus (A^p)^2, \quad \text{and} \quad f(s) = f(t) \implies s = t.$$

To verify (G1), suppose that two vertices of $H^q_0 \times H^q_1$ have the same f-color. If either of the corresponding ordered pairs is new, (F) says that the two ordered pairs are identical, so they cannot be adjacent. If both pairs belong to $(A^p)^2$, use (G1) for p.

We verify (G2). By symmetry between the two coordinates—interchanging H_0 and H_1 and transposing f—it is enough to consider $e = 0$. Fix a witnessing pair of synchronized walks for $(a, b) \in \mathsf{Sync}^{x,0}_q$, and suppose, toward a contradiction, that

$$h(\{u, v\}) = 1, \quad \min\{u, v\} < x, \quad f(a, u) = f(b, v). \tag{4.1}$$

Property (F) and $u \neq v$ imply that the ordered pairs (a, u) and (b, v) both belong to $(A^p)^2$. In particular $a, b, u, v \in A^p$.

Let $a_0, \dots, a_k$ and $b_0, \dots, b_k$ be the two synchronized walks. By synchronization,

$$f(a_j, a_{j+1}) = f(b_j, b_{j+1}) \qquad (j < k).$$

If either (a_j, a_{j+1}) or (b_j, b_{j+1}) is new, (F) implies $(a_j, a_{j+1}) = (b_j, b_{j+1})$. If $a = b$, then $(a, a) \in \mathsf{Sync}_p^{a,0}$ and $\min\{u, v\} < x \le a$. Thus (G2) for p, applied with base point a, gives $f(a, u) \neq f(a, v)$, whereas (4.1), since $a = b$, gives $f(a, u) = f(a, v)$, a contradiction. Assume that $a \neq b$, and let j be the last index at which the two walks agree. Every step from index j onward is old, and all vertices in this terminal part of the walks belong to A^p. Let z be the least ordinal occurring in this terminal part. If $z = a_j = b_j$, use the two terminal sequences. Otherwise use

$$z, a_j, a_{j+1}, \dots, a_k = a \quad \text{and} \quad z, b_j, b_{j+1}, \dots, b_k = b.$$

The first steps of these two sequences are identical, and the remaining corresponding steps have the same f^p-colors. Thus in either case

$$(a, b) \in \mathsf{Sync}_p^{z,0}.$$

All vertices of the terminal part are at least z, while $\min\{u, v\} < x \le z$. Now (G2) for p again contradicts (4.1). □

We next isolate the amalgamation used both for the chain condition and for the chromatic-number argument.

Definition 4.5 (Twins and separated twins)**.** Two good preconditions p and q are called *twins* if $|A^p| = |A^q|$ and the increasing bijection $\sigma : A^p \to A^q$ fixes $R = A^p \cap A^q$ pointwise and is an isomorphism, that is,

$$h^q(\{\sigma(a), \sigma(b)\}) = h^p(\{a, b\}), \qquad f^q(\sigma(a), \sigma(b)) = f^p(a, b).$$

Write $D_p = A^p \setminus R$ and $D_q = A^q \setminus R$. The twins are *separated* if

$$R < D_p < D_q,$$

where $X < Y$ means that every member of X is smaller than every member of Y.

Lemma 4.6 (Amalgamation lemma)**.** *Suppose that p and q are separated twins and that $D_p \neq \varnothing$. Let $x = \min D_p$. For each $i < 2$ there is a good precondition $r \le p, q$ such that*

$$h^r(\{x, \sigma(x)\}) = i.$$

Proof. We prove the case $i = 0$; the case $i = 1$ is analogous.

Put $A^r = A^p \cup A^q$, and define the projection $\pi : A^r \to A^p$ by

$$\pi \restriction A^p = \mathrm{id}_{A^p}, \qquad \pi \restriction D_q = \sigma^{-1}.$$

Let

$$\mathcal{F} = \big\{\{a, b\} : a \in D_p, b \in D_q, (a, \pi(b)) \in \mathsf{Sync}_p^{x,0}\big\}. \tag{4.2}$$

On $[A^p]^2 \cup [A^q]^2$ let h^r agree with $h^p \cup h^q$. For $a \in D_p$ and $b \in D_q$, put

$$h^r(\{a, b\}) = \begin{cases} 0, & \{a, b\} \in \mathcal{F}, \\ h^p(\{a, \pi(b)\}), & \{a, b\} \notin \mathcal{F}. \end{cases} \tag{4.3}$$

Formula (4.3) is well-defined. Indeed, if $a = \pi(b)$, the identical walk from x to a puts $\{a, b\}$ in $\mathcal{F}$, so the first line applies. Otherwise $\{a, \pi(b)\} \in [A^p]^2$, and the second line is defined.

Let f^r extend $f^p \cup f^q$, and give all ordered pairs in

$$(A^r)^2 \setminus \big((A^p)^2 \cup (A^q)^2\big) \tag{4.4}$$

pairwise distinct values outside $\mathrm{ran}(f^p)$. The functions f^p and f^q agree on their common domain, so f^r is well-defined. Call the ordered pairs in (4.4) *new* and all

other ordered pairs *old*. Since p and q are twins, $\operatorname{ran}(f^q) = \operatorname{ran}(f^p)$; hence f^r has the analogous freshness property:

$$(F_r)\qquad \begin{aligned} &s \in (A^r)^2,\\ &t \in (A^r)^2 \setminus \big((A^p)^2 \cup (A^q)^2\big), \quad \text{and} \quad f^r(s) = f^r(t) \implies s = t. \end{aligned}$$

Set $r = (A^r, h^r, f^r)$. Since $(x,x) \in \mathsf{Sync}_p^{x,0}$, (4.2)–(4.3) give $h^r(\{x, \sigma(x)\}) = 0$.

We first prove (G1). Suppose otherwise. Thus, for suitable $a_0, a_1 \in A^r$ and $b_0, b_1 \in A^r$,

$$h^r(\{a_0, b_0\}) = 0, \qquad h^r(\{a_1, b_1\}) = 1, \qquad f^r(a_0, a_1) = f^r(b_0, b_1). \tag{4.5}$$

If either ordered pair in (4.5) is new, (F_r) forces the two ordered pairs to be identical, which is impossible. If both ordered pairs belong to $(A^p)^2$, or both belong to $(A^q)^2$, use (G1) in that condition. We may therefore assume, after interchanging the two product vertices if necessary, that $(a_0, a_1) \in (A^p)^2$ and $(b_0, b_1) \in (A^q)^2$. Put $b'_j = \pi(b_j)$ for $j < 2$. Then

$$f^p(a_0, a_1) = f^p(b'_0, b'_1). \tag{4.6}$$

For $j < 2$, call the edge $\{a_j, b_j\}$ *special* if $\{a_j, b_j\} \in \mathcal{F}$. A nonspecial edge has the same color as the genuine edge $\{a_j, b'_j\}$ of p; its projected endpoints are distinct, since equality would make the edge special by the diagonal observation above. If neither edge is special, (4.5)–(4.6) contradict (G1) for p. Every special edge has color 0, so the only remaining possibility is that the 0-th edge is special. Hence

$$(a_0, b'_0) \in \mathsf{Sync}_p^{x,0}, \qquad h^p(\{a_1, b'_1\}) = 1. \tag{4.7}$$

Let

$$c_0, \dots, c_k = a_0 \quad \text{and} \quad d_0, \dots, d_k = b'_0$$

be synchronized walks from x witnessing the first assertion in (4.7). If $a_1, b'_1 \geq x$, extend these walks by appending a_1 to the first and b'_1 to the second. By (4.6), the two added steps have the same f^p-color. The extended walks therefore witness $(a_1, b'_1) \in \mathsf{Sync}_p^{x,0}$. Since $b'_1 = \pi(b_1)$, (4.2) gives $\{a_1, b_1\} \in \mathcal{F}$; hence the other edge is also special, a contradiction. Otherwise $\min\{a_1, b'_1\} < x$, and (G2) for p, applied to (4.7), contradicts (4.6). Thus (G1) holds in r.

It remains to prove (G2). Recall that $x = \min D_p$ is fixed throughout the construction; we use y for the arbitrary base point occurring in (G2). It is enough to consider $e = 0$; the case $e = 1$ is analogous. Suppose that

$$(a, b) \in \mathsf{Sync}_r^{y,0}, \qquad h^r(\{u, v\}) = 1, \qquad \min\{u, v\} < y, \tag{4.8}$$

but

$$f^r(a, u) = f^r(b, v). \tag{4.9}$$

By (F_r) and $u \neq v$, the ordered pairs (a, u) and (b, v) are both old: otherwise (4.9) would make them identical and hence give $u = v$. Let $a_0, \dots, a_k = a$ and $b_0, \dots, b_k = b$ be synchronized walks from y witnessing the first assertion in (4.8). For every $j < k$,

$$f^r(a_j, a_{j+1}) = f^r(b_j, b_{j+1}).$$

If either (a_j, a_{j+1}) or (b_j, b_{j+1}) is new, (F_r) implies that these two ordered pairs are identical.

There are three cases. If $y \in R$, project both witnessing walks by π. Fix $j < k$. If both (a_j, a_{j+1}) and (b_j, b_{j+1}) are old, then the definition of π and the fact that p and q are twins give

$$f^p(\pi(a_j), \pi(a_{j+1})) = f^p(\pi(b_j), \pi(b_{j+1})).$$

If either ordered pair is new, the preceding paragraph shows that the two ordered pairs, and hence their projections, are identical. Thus the displayed equality holds for every $j < k$. All projected vertices are at least y, so

$$(\pi(a), \pi(b)) \in \mathsf{Sync}_p^{y,0}.$$

The edge $\{u, v\}$ cannot join D_p to D_q, since both sets lie above every member of R, whereas $\min\{u, v\} < y \in R$. Consequently

$$h^p(\{\pi(u), \pi(v)\}) = 1, \qquad \min\{\pi(u), \pi(v)\} < y.$$

Moreover, projecting (4.9) gives

$$f^p(\pi(a), \pi(u)) = f^p(\pi(b), \pi(v)).$$

Together with $(\pi(a), \pi(b)) \in \mathsf{Sync}_p^{y,0}$, these statements contradict (G2) for p.

If $y \in D_q$, every vertex of the witnessing walks lies in D_q. Thus the first assertion in (4.8) already holds in q. Oldness of the two pairs in (4.9) implies $u, v \in A^q$, and (G2) for q gives the contradiction.

Finally suppose that $y \in D_p$. An induction on $j \leq k$ shows that either $a_j = b_j$, or $a_j, b_j \in D_p$, or $a_j, b_j \in D_q$. Indeed, all vertices of the walks lie above R and hence belong to $D_p \cup D_q$. If (a_j, a_{j+1}) is old, then a_j and a_{j+1} belong to the same one of D_p, D_q; the analogous statement holds for (b_j, b_{j+1}). If either ordered pair is new, (F_r) makes the two ordered pairs identical. It follows from (4.9) that a, b, u, v all lie in the same support, either A^p or A^q. Let $s \in \{p, q\}$ be the corresponding condition. Thus $\{u, v\}$ and equality (4.9) are inherited by s. If $a = b$, apply (G2) in s with base a, noting that $\min\{u, v\} < y \leq a$. Suppose now that $a \neq b$, and let

$$j = \max\{\ell \leq k : a_\ell = b_\ell\}.$$

Then $j < k$. For every $j \leq \ell < k$, both $(a_\ell, a_{\ell+1})$ and $(b_\ell, b_{\ell+1})$ are old; otherwise (F_r) would make the two ordered pairs identical, contradicting the choice of j. It follows from the induction above that all vertices in the two terminal sequences

$$a_j, a_{j+1}, \ldots, a_k = a \quad \text{and} \quad b_j, b_{j+1}, \ldots, b_k = b$$

belong to A^s. Let

$$z = \min\{a_\ell, b_\ell : j \leq \ell \leq k\}.$$

If $z = a_j = b_j$, use the two terminal sequences displayed above. Otherwise use

$$z, a_j, a_{j+1}, \ldots, a_k = a \quad \text{and} \quad z, b_j, b_{j+1}, \ldots, b_k = b.$$

The first steps of these two sequences are identical, and the remaining corresponding steps have equal f^s-colors. Thus in either case the resulting sequences are synchronized walks in A^s witnessing

$$(a, b) \in \mathsf{Sync}_s^{z,0}.$$

Since $z \geq y$ and $\min\{u, v\} < y$, (G2) for s contradicts (4.9). This proves (G2), and hence r is good. □

Proof of Theorem 1.3. We verify that P has the properties asserted in the theorem. The empty precondition is good. Lemma 4.4 shows that, for every $\alpha < \omega_1$, the set

$$\{p \in P : \alpha \in A^p\} \tag{4.10}$$

is dense. It also shows that, for every $n < \omega$ and $i < 2$, it is dense to have an n-element subset of A^p all of whose pairs receive the h^p-color i.

Clearly $|P| = \omega_1$. We check the c.c.c. Given an uncountable family of conditions, first thin it so that its supports are distinct; this is possible because only countably many conditions have any one fixed finite support. Apply the Δ-system lemma to the supports. After thinning, they have a common finite root R and pairwise disjoint, nonempty petals. Only countably many petals can meet $\max(R) + 1$, so

we may discard them and assume that R lies below every petal (with the evident interpretation when $R = \varnothing$). There are only countably many possibilities for the sizes of the supports and for the finite h- and f-matrices. We may therefore thin again so that every increasing bijection between two supports fixes R and is an isomorphism. Fix one petal D_p. Only countably many of the other pairwise disjoint petals meet $\max(D_p)+1$, so one may choose a petal D_q with $D_p < D_q$. The corresponding conditions are separated twins, so Lemma 4.6 gives a common extension. Thus P is c.c.c.

Let $K \subseteq P$ be generic. By (4.10), the unions

$$h = \bigcup_{p\in K} h^p : [\omega_1]^2 \longrightarrow 2, \qquad f = \bigcup_{p\in K} f^p : \omega_1^2 \longrightarrow \omega$$

are total. Put $E_i = h^{-1}\{i\}$ and $H_i = (\omega_1, E_i)$. Then $H_1 = \overline{H_0}$. Requirement (G1) implies that f is a proper coloring of $H_0 \times H_1$, so

$$\operatorname{chr}(H_0 \times H_1) \leq \omega. \tag{4.11}$$

The dense sets mentioned after (4.10) ensure that each H_i contains K_n for every $n < \omega$. Pairing a copy of K_n in H_0 with a copy in H_1 embeds K_n into $H_0 \times H_1$. Hence equality holds in (4.11).

We finish by proving that $\operatorname{chr}(H_i) = \omega_1$ for $i < 2$. Suppose that some $p_* \in P$ forces that $\dot c : \omega_1 \to \omega$ is a proper coloring of H_i. For each α in the stationary set of nonzero countable limit ordinals, choose $p_\alpha \leq p_*$ such that $\alpha \in A^{p_\alpha}$ and p_α decides $\dot c(\alpha)$. The map

$$\alpha \longmapsto \begin{cases} 0, & A^{p_\alpha} \cap \alpha = \varnothing, \\ \max(A^{p_\alpha} \cap \alpha) + 1, & A^{p_\alpha} \cap \alpha \neq \varnothing \end{cases}$$

is regressive. By the pressing-down lemma it is constant on a stationary set. On that set the finite sets $A^{p_\alpha} \cap \alpha$ are all contained in one countable ordinal; there are only countably many possibilities for them. Thus, after thinning, they are a fixed finite set L.

Apply the Δ-system lemma to the supports and thin once more. Its root must equal L: after discarding boundedly many indices, every root element lies below α, while every member of L belongs to every support. Thinning by the countably many finite matrix types and by the countably many values of $\dot c(\alpha)$, we may arrange that

(1) the A^{p_α} have the common root $R = L$ and form pairwise isomorphic conditions;
(2) $R < A^{p_\alpha} \setminus R$, and $\alpha = \min(A^{p_\alpha} \setminus R)$;
(3) all the conditions force the same value, say m, for $\dot c(\alpha)$.

Choose $\alpha < \beta$ in the remaining uncountable set so that the two petals are successive blocks. Their increasing isomorphism sends α to β. Lemma 4.6, used with the prescribed color i, gives $r \leq p_\alpha, p_\beta$ such that $h^r(\{\alpha,\beta\}) = i$. But

$$r \Vdash \dot c(\alpha) = \dot c(\beta) = m,$$

contrary to the asserted properness of $\dot c$. Thus neither H_i is countably chromatic. Since P is c.c.c. and each H_i has vertex set ω_1, we have $\operatorname{chr}(H_0) = \operatorname{chr}(H_1) = \omega_1$. Taking $G = H_0$ completes the proof. □

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author used ChatGPT to correct grammatical and typographical errors and misprints. All mathematical arguments were

created by the author. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the publication.

References

[1] J. Cummings, T. Eisworth, and J. T. Moore, *On minimal non-σ-scattered linear orders*, Adv. Math. **441** (2024), article 109540, doi:10.1016/j.aim.2024.109540.

[2] A. Hajnal, *The chromatic number of the product of two $\aleph_1$-chromatic graphs can be countable*, Combinatorica **5** (1985), no. 2, 137–139, doi:10.1007/BF02579376.

[3] S. T. Hedetniemi, *Homomorphisms of graphs and automata*, University of Michigan Technical Report 03105–44–T, 1966.

[4] J. T. Moore, *ω_1 and $-\omega_1$ may be the only minimal uncountable linear orders*, Michigan Math. J. **55** (2007), no. 2, 437–457, doi:10.1307/mmj/1187647002.

[5] E. A. Nordhaus and J. W. Gaddum, *On complementary graphs*, Amer. Math. Monthly **63** (1956), 175–177, doi:10.2307/2306658.

[6] S. Poljak and V. Rödl, *On the arc-chromatic number of a digraph*, J. Combin. Theory Ser. B **31** (1981), 190–198, doi:10.1016/S0095-8956(81)80024-X.

[7] A. Rinot, *Hedetniemi's conjecture for uncountable graphs*, J. Eur. Math. Soc. **19** (2017), no. 1, 285–298, doi:10.4171/JEMS/666.

[8] N. Sauer, *Hedetniemi's conjecture—a survey*, Discrete Math. **229** (2001), 261–292, doi:10.1016/S0012-365X(00)00213-2.

[9] R. Shalev, *More minimal non-σ-scattered linear orders*, Eur. J. Math. **10** (2024), article no. 74, doi:10.1007/s40879-024-00780-y.

[10] Y. Shitov, *Counterexamples to Hedetniemi's conjecture*, Ann. of Math. (2) **190** (2019), no. 2, 663–667, doi:10.4007/annals.2019.190.2.6.

[11] L. Soukup, *On chromatic number of product of graphs*, Comment. Math. Univ. Carolin. **29** (1988), no. 1, 1–12.

[12] X. Zhu, *A survey on Hedetniemi's conjecture*, Comput. Sci. Rev. **58** (2025), Article 100794, doi:10.1016/j.cosrev.2025.100794.